



The Likely Thickness of Europa's Icy Shell

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Abstract

The icy outer shell of Jupiter's moon Europa is believed to be geologically active, and geological processes within the ice may be key to sustaining chemical disequilibria and potentially habitable environments within the icy shell and ocean. Currently, there are insufficient observational data to determine the thickness of the icy shell, obfuscating crucial aspects of interior processes. Although a range of icy shell thicknesses ($\sim 10^{-1}$ – 10^2 km) have been proposed, not all are equally likely given the current knowledge of Europa. In this study, I construct a probability density distribution for Europa's present-day mean icy shell thickness using a heat flux balance, assuming a simple two-layered model consisting of a partially porous conductive upper layer, and a ductile interior layer that may be experiencing solid-state convection. Current best estimate (CBE) values and uncertainties for ~ 20 key parameters are propagated through this steady-state heat flux balance to construct $\sim 10^7$ independent simulations. Additionally, I estimate present-day values for the icy shell layer thicknesses, the rocky interior and icy heat fluxes, icy shell thermal properties, and convective ice viscosity. I predict Europa's total icy thickness to be ~ 23 – 47 km, with a CBE thickness of $24.3^{+22.8}_{-1.5}$ km. This result is highly skewed; icy shells many tens of kilometers thick are much more probable than icy shells < 10 km thick. The resulting distributions have key implications for future spaceborne, landed, and subsurface exploration of Europa, as well as studies of icy shell processes, thermal and mechanical state, and habitability across the icy ocean worlds.

Unified Astronomy Thesaurus concepts: [Europa \(2189\)](#); [Astrobiology \(74\)](#); [Tectonics \(2175\)](#)

1. Introduction

Europa, an icy Galilean satellite of Jupiter, has risen to the forefront of planetary exploration and the search for habitable environments within our solar system. Europa is the primary target of NASA's planned flagship-class Europa Clipper mission (Howell & Pappalardo 2020), one of three ocean world targets of ESA's L-Class Jupiter Icy Moons Explorer (JUICE) mission (Grasset et al. 2013), and an extended mission target for NASA's New Frontiers-class Juno mission (Bolton et al. 2017).

The icy surface of Europa's outer shell exhibits one of the youngest average surface ages in the solar system (~ 40 – 90 Myr; e.g., Zahnle et al. 2003), requiring recent or current geologic resurfacing (Figure 1). Visible surface deformation occurs along convergent boundaries (e.g., Kattenhorn & Prockter 2014), strike-slip boundaries (e.g., Nimmo & Gaidos 2002; Hammond 2020), extensional boundaries (e.g., Howell & Pappalardo 2018), and through processes that produce exotic and chaotic terrains (e.g., Doggett et al. 2009). These resurfacing processes may convey material between the surface and interior ocean, influencing chemical disequilibria within the interior and the habitability of Europa (Hand et al. 2007).

The thickness of the icy shell, and therefore the potential for geologic activity and material exchange, is underconstrained by current data. Following discoveries at Europa by the Galileo mission, Billings & Kattenhorn (2005) addressed the growing "Great Thickness Debate." That work cataloged dozens of prior

studies, collating estimates for full icy shell thicknesses of a few hundred meters to many tens of kilometers.

Importantly, Billings & Kattenhorn (2005) note that the myriad processes leveraged to infer icy shell thickness are sensitive to different regions of the icy shell and to different timescales. For example, flexural studies are most sensitive to the portion of the cold, near-surface ice that behaves elastically on the timescale of the process studied, while structural analyses are most sensitive to the portions of the shell that experience brittle failure in response to geologic stresses. Studies of Europa may also sample past icy shell thicknesses at the time a process was active, but that no longer reflect the present state. The thickest icy shells were predicted by analyses of the potential thermal state, which are less sensitive to the time and energy scales of transient processes like impacts or brittle failure. The debate persists, with recent estimates offering a wide range of bounds on shell thickness, including > 3 km from rheological studies of landforms (Leonard et al. 2020), 5–30 km from thermodynamic analyses (Quick & Marsh 2015; Green et al. 2021), up to 50–90 km from geodynamic numerical simulations (Vilella et al. 2020).

The present study provides a basis for the understanding that not all predictions of icy shell thickness are equally likely. Here, I estimate a probability distribution for Europa's present-day mean equatorial icy shell thickness by solving a heat flux balance across the iron-rich core, rocky interior, ocean, and icy shell. The resulting probability density distribution describes the relative likelihood of a particular thickness occurring given the underlying uncertainties in our knowledge of Europa. This study assumes no significant present-day changes in icy shell thickness, which would introduce unbounded latent heat lost or gained through the changing phase of water. As a result, I describe how deviations of future measurements from the steady-state predictions modeled here may be used to investigate transient icy shell behavior. For simplicity, this



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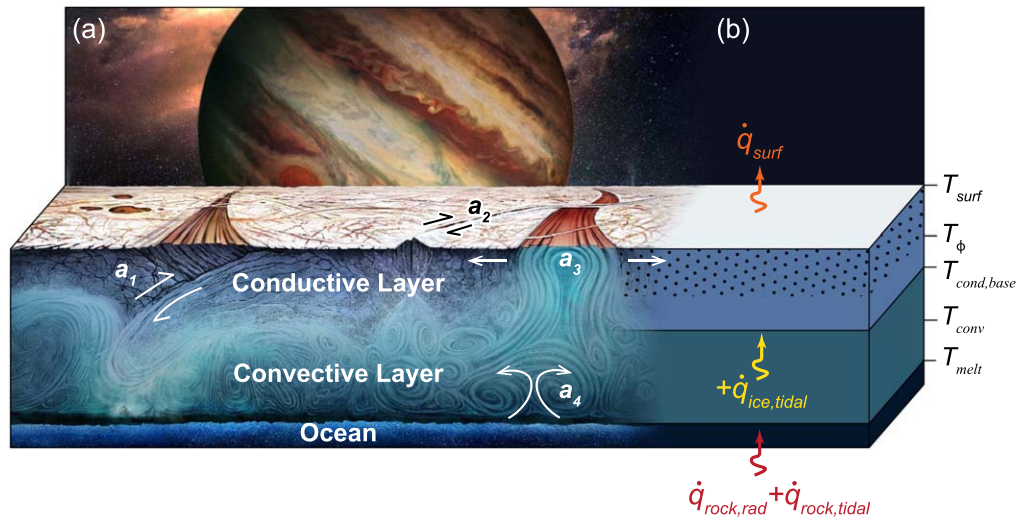


Figure 1. Schematic of Europa’s icy shell showing the conductive and convective layers considered in this study. (a) An artistic illustration of the icy shell demonstrates how geodynamic processes may result in regional thickness variations in the layered ice structure, including (a_1) thickening of the conductive layer beneath convergent margins such as “subsumption zones” (e.g., Kattenhorn & Prockter 2014), (a_2) thinning of the conductive layer due to frictional shear heating along strike-slip margins (e.g., Nimmo & Gaidos 2002; Hammond 2020) and (a_3) extensional band formation (e.g., Howell & Pappalardo 2018), and short-scale lateral variations in temperature due to (a_4) solid-state convection within the icy shell. (b) A simplified schematic of the icy shell as used in this study. Radiogenic ($\dot{q}_{\text{rock,rad}}$) and tidal ($\dot{q}_{\text{rock,tidal}}$) heat generated in the rocky interior enters the base of the icy shell, is augmented by the tidal heat generated in the convective layer ($\dot{q}_{\text{ice,tidal}}$), conducts through the upper icy shell, and radiates from the surface to space ($\dot{q}_{\text{surf}}$). Black dots represent porosity, and the relevant interface temperatures (T) are labeled as in Figure 2.

study does not consider the effect of latitude or longitude on tidal heat dissipation, which may result in regional variations in icy shell thickness (Ojakangas & Stevenson 1989) that could be investigated in a future study.

2. Model Design

2.1. Heat Flux Balance

Europa is thought to be fully differentiated (Anderson et al. 1998; Kuskov & Kronrod 2005), consisting of an iron-rich core entombed in a silicate body and draped in a predominantly H_2O hydrosphere. Heat is produced within the rocky interior through radiogenic heating ($\dot{q}_{\text{rock,rad}}$) and from the dissipation of tidal energy as heat ($\dot{q}_{\text{rock,tidal}}$; e.g., Barr & Showman 2009; Sotin et al. 2009; Hussmann et al. 2016). While some tidal energy is dissipated in the ocean, it is neglected here because the magnitude of ocean dissipation from both eccentricity tides ($\sim 10^{-8} \text{ mW m}^{-2}$) and obliquity tides ($\sim 10^{-1} \text{ mW m}^{-2}$) are predicted to be small (Beuthe 2016; Hay & Matsuyama 2019) compared to the predicted magnitude of radiogenic heating ($\sim 10^1 \text{ mW m}^{-2}$; Spohn & Schubert 2003; Barr & Showman 2009; Sotin et al. 2009).

Covering the ocean, Europa’s icy shell is thought to be composed of two primary layers (Figure 1). The inner layer consists of a warm, ductile asthenosphere that may be convecting (e.g., Pappalardo et al. 1998; Barr & McKinnon 2007), relaxing stresses through viscous flow before they build to cause brittle failure. This ductile layer is predicted to dissipate significant amounts of tidal energy as heat ($\dot{q}_{\text{ice,tidal}}$), potentially serving as the most significant contributor to the global heat flux balance (Tobie et al. 2003; Mitri & Showman 2005; Barr & Showman 2009; Sotin et al. 2009).

In contrast, the outermost icy layer includes the cold, brittle, and elastic icy lithosphere. Only tidal heat generation within the convective layer is considered because viscosities in the colder conductive layer are much higher, inhibiting the

response of the ice to rapid tidal forcing and resulting in a minority contribution to dissipated heat (Tobie et al. 2003; Sotin et al. 2009). I assume that heat entering the base of this outermost layer is conveyed to the surface by conduction, where it is radiated to space at a rate $\dot{q}_{\text{surface}}$.

For simplicity, this study will use the terms “convective layer” and “conductive layer,” referring to these complex regions by their dominant mechanism of global-scale heat transfer. The modeled conductive layer is porous to a specified depth. The convective layer is approximated as isothermal for simplicity, assuming that the thin thermal boundary layers that develop at the edges of convection cells participate in dissipation. A schematic of the icy shell thermal profile is shown in Figure 2. For some models that produce implausibly thin convective layers, a conductive basal layer is substituted for the convective layer as described below.

This study relies on the assumption that the heat flux through Europa is in steady state, producing the following heat flux balance:

$$\dot{q}_{\text{surface}} = \dot{q}_{\text{rock,rad}} + \dot{q}_{\text{rock,tidal}} + \dot{q}_{\text{ice,tidal}}. \quad (1)$$

To investigate the dissipation of tidal energy within the icy shell, I compute the time-averaged dissipation rate for an incompressible Maxwell solid subject to sinusoidal strain variations of magnitude ϵ_0 over the European orbital frequency ω (Tobie et al. 2003; Mitri & Showman 2005; Nimmo et al. 2007; Sotin et al. 2009):

$$\dot{q}_{\text{ice,tidal}} = D_{\text{conv}} \frac{\eta_c \epsilon_0^2 \omega^2}{1 + \eta_c^2 \omega^2 / \mu_{E,\text{ice}}^2}. \quad (2)$$

Here, η_c is the viscosity of the convecting ice layer, $\mu_{E,\text{ice}}$ is the layer rigidity, and D_{conv} is the layer thickness.

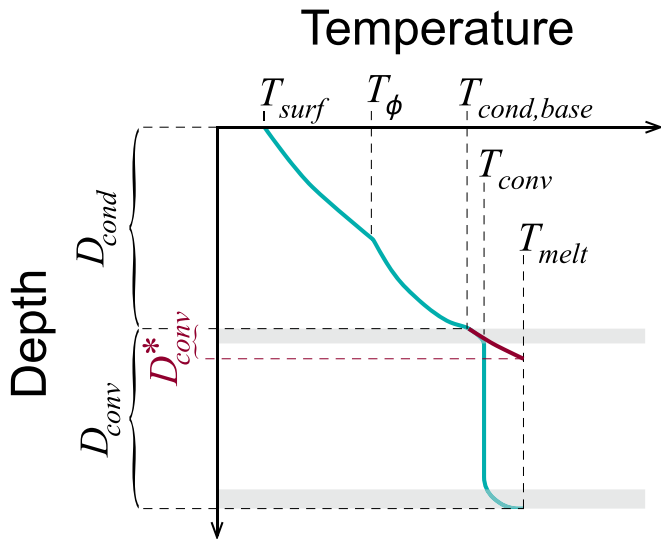


Figure 2. Schematic thermal profile showing a temperature-, porosity-, and composition-dependent conductive thermal profile between the surface temperature (T_{surf}), and the porosity curing temperature (T_{ϕ}); a temperature- and composition-dependent conductive thermal profile extending beneath the porous layer to the base of the conductive layer (D_{cond}) at temperature $T_{cond,base}$; and a convective (blue) thermal profile maintaining an average isothermal temperature (T_{conv}) to the base of the convective layer D_{conv} . The thermal boundary layers associated with convection (gray bars) are not simulated, but assumed to be part of the isothermal convecting region for simplicity. The red line and D_{conv}^* depict an uncommon special case where the model predicts a warm, ductile lower layer that is nonphysically thin. In this case, the convective layer is replaced with a conductive layer spanning from a depth D_{cond} to the base of the icy shell. For simplicity and consistency in reporting, this special conductive case retains the label D_{conv} throughout the paper.

The radiogenic heat produced by the rocky core is a function of the rocky body mass and primordial composition:

$$\dot{q}_{rock,rad} = \frac{H_{rad}m}{A_{ice}}, \quad (3)$$

where H_{rad} is the heat produced per unit mass, m is the mass of the solid body, and A_{ice} is the surface area of the icy shell where the heat flux is evaluated.

To what degree Europa's rocky interior dissipates tidal energy is unknown (e.g., O'Brien et al. 2002; Moore et al. 2009; Běhounková et al. 2020). Generally the Maxwell model shown above (Equation (2)) is not appropriate for characterizing heat produced in highly dissipative silicate interiors (e.g., Sotin et al. 2009). Orbiting Jupiter in concert with Europa is Io, the most volcanically active terrestrial body in the solar system. Io's volcanic activity is supported by strong tidal dissipation within the satellite (Lainey et al. 2009). Sotin et al. (2009) summarize the argument that Europa should have a less dissipative silicate core than Io because Europa is not as efficient at elevating global surface heat flux in response to frictional tidal dissipation, resulting in a potentially cooler rocky interior that would dissipate much less heat than occurs through radioisotope decay. That study notes that suppressed warming acts to stymie the rheological feedbacks that can lead to runaway silicate tidal heat dissipation and allow Europa to experience Io-like dissipation. Additionally, the decoupling of Europa's icy shell and rocky interior by the ocean should act to reduce silicate dissipation (Tobie et al. 2005).

The dissipative nature of a body in response to tidal forcing can be characterized by a global dissipation factor, Q , and the second-degree Love number k_2 , which describes how elastic

energy is partitioned to deform and heat a body. The ratio between the stored peak elastic energy and the energy dissipated during one tidal cycle is (Segatz et al. 1988)

$$Q = \frac{k_2}{\text{Im}(k_2)}, \quad (4)$$

where $\text{Im}(k_2)$ is the imaginary part of k_2 . The total dissipated power is given by

$$P_{\text{tidal}} = \frac{21}{2} \frac{|k_2|}{Q} \frac{\omega^5 R^5}{G} e^2. \quad (5)$$

Here, R is the radius, G is the universal gravitational constant, and e is the eccentricity.

A value of k_2/Q that corresponds to that of Io (0.015 ± 0.003 ; Lainey et al. 2009) would result in a mean total global dissipation within Europa (using the parameters of Table 1) of ~ 8 TW. Recently, Běhounková et al. (2020) explore heating and melting within the silicate interior using three-dimensional models of tidal dissipation. Considering a wide range of input parameters, they find that the present-day dissipation within the interior of Europa during its recent past is likely to be < 100 GW with limited radiogenic heating, and peak dissipation in the recent past is unlikely to exceed ~ 500 GW at much higher eccentricities. In all cases, for the present-day eccentricity, Europa is likely to dissipate < 200 GW in the silicate interior due to tidal heating. Thus, the most sophisticated characterizations of dissipation within Europa's interior fail to reproduce Io-like behavior.

Direct application of the results of Běhounková et al. (2020) to this study are nuanced, as they may not capture the full range of potential interior conditions or physics. For example, fully incorporating the dissipation associated with orbital phase lag on Europa may result in substantially increased silicate dissipation (Hussmann et al. 2016). Additionally, the models of Běhounková et al. (2020) were not systematically applied in a way that would preserve information about the underlying distributions of the input parameters.

Therefore, to capture the best present understanding of Europa's interior dissipation and calculate a value of $\dot{q}_{rock,tidal}$, I assume a conservative distribution in tidally dissipated power, P_{tidal} , that is informed by the sophisticated and broad analysis of Běhounková et al. (2020), while preserving a small but nonzero probability of Io-like dissipation, as further discussed in Section 3.1. Thus,

$$\dot{q}_{rock,tidal} = \frac{P_{\text{tidal}}}{A_{ice}}. \quad (6)$$

To complete the heat flux balance, the flux through the conductive layer of the icy shell, $\dot{q}_{ice,cond}$, is set equal to the heat flux out of the surface,

$$\begin{aligned} \dot{q}_{\text{surface}} &= \dot{q}_{ice,cond} \\ &= \bar{k}(T, \phi, f_s) \frac{T_{cond,base} - T_{surf}}{D_{cond}}, \end{aligned} \quad (7)$$

where $\bar{k}(T, \phi, f_s)$ is the effective thermal conductivity averaged across variations in temperature, T , porosity, ϕ , and salt fraction, f_s ; $T_{cond,base}$ is the temperature at the base of the conductive layer; T_{surf} is the surface temperature; and D_{cond} is the conductive layer thickness.

As described below, I derive and sample various underlying distributions as inputs to Equations (1)–(7) to obtain self-

Table 1
Standard Values for Europa Parameters

Description	Symbol	Value	Units	References
Radius	r_E	1561	km	JPL (2021)
Mass	m_E	4.80×10^{22}	kg	JPL (2021)
Orbital frequency	ω	2.047×10^{-5}	s^{-1}	JPL (2021)
Eccentricity	e	0.00981	...	JPL (2021)
Gravitational acceleration	g	1.315	$m s^{-2}$	JPL (2021)
Ice molar volume	V_m	1.97×10^{-5}	m^3	Goldsby & Kohlstedt (2001), Fletcher (2009)
Melting temperature	$T_{melt,0}$	273.13	K	e.g., Spohn & Schubert (2003)

consistent values for the layer thicknesses D_{cond} and D_{conv} , which sum to the total icy shell thickness D_{ice} .

2.2. Model Approach

The general approach of this study is to construct a distribution of icy shell thicknesses for Europa by generating many ($\sim 10^7$) possible European icy shells. Standard values used in this study are listed in Table 1. For the input parameters with significant uncertainty, various distributions are selected that capture the current best estimates for the values as well as the associated uncertainties and confidence levels (Table 2).

The wide range of distributions and nonlinear relationships of the underlying variables makes deriving the analytical distributions for layer and total icy shell thicknesses prohibitive. Therefore, I choose to employ a Monte Carlo approach, sampling each distribution independently to build a spectrum of possible Europas. In this context, sampling is performed by prescribing a distribution of probabilities associated with the range of values of interest, and then selecting the value of the variable at random according to the prescribed distribution. The full Python Monte Carlo model software is provided as a GitHub repository (Howell 2021).

The general structure of the Monte Carlo is as follows:

1. Construct a distribution to sample the rocky interior heat flux;
2. Determine the thermal properties of the icy conductive layer;
3. Bound the range of potential conductive layer thicknesses;
4. Sample conductive thickness and calculate the corresponding convective and total icy shell thicknesses.

To determine current best estimate (CBE) values, histograms of the Monte Carlo outputs are normalized to produce differential probability density distributions. The CBE value is taken as the peak of the differential probability density distribution—analogueous to the most likely value to occur for Europa—and the uncertainties highlight resulting asymmetries in these distributions.

The resulting Monte Carlo distributions are then filtered through a third-order Savitzky Golay filter (Press & Teukolsky 1990) with a window size of 1/20 of the total histogram signal length to smooth out random noise and identify the peak. Uncertainties are reported at the one standard deviation (1σ) level. I extract CBE values and uncertainties from conditional discrete probability distributions evaluated at the CBE total icy shell thickness ($D_{ice,CBE}$), i.e., I report $P(x|D_{ice,CBE})$ for some distribution x . The CBE values and uncertainties for the parameters derived from the Monte Carlo model are reported in Table 3.

3. Model Method

3.1. Step 1: Rocky Interior Heat Flux

In this first step, I construct and sample a distribution for heat flux originating from the rocky and metallic interior of Europa (Figure 3(a)). After Spohn & Schubert (2003), I consider the specific heat production for CV (4.5 pW kg^{-1}) and CO (4 pW kg^{-1}) type chondrites, as well as inferred values for the Earth and Moon (4 pW kg^{-1}). I assume a normally distributed value for radiogenic specific heat production, H_{rad} , with mean 4.5 pW kg^{-1} and standard deviation $\sigma = 1.0 \text{ pW kg}^{-1}$ (Spohn & Schubert 2003; Hussmann et al. 2010).

To bound the tidal heat generated within the silicate interior as discussed above, I apply a conservative distribution informed by recent modeling efforts (Běhounková et al. 2020). Given that dissipation is controlled by material viscosity, which varies exponentially with temperature, I employ a log-normal distribution for dissipated power within the silicate interior due to diurnal tides, $P_{rock,tidal}$. I assume a mean of 100 GW (Běhounková et al. 2020) and 3σ uncertainty of one order of magnitude. This results in a distribution of dissipated powers consistent with recent research, and that allows for the low-likelihood occurrence of Io-like dissipation.

Using Equations (3) and (6) to obtain $\dot{q}_{rock,rad}$ and $\dot{q}_{rock,tidal}$, respectively, a distribution is obtained for the total heat flux from the solid interior (Figure 3(a)). The primary contributor to solid interior heat flux is radiogenic heating, with the minority contribution resulting from tidal heating. The CBE interior heat flux is $24.5_{-10.8}^{+3.9} \text{ mW m}^{-2}$ (Figure 3; Table 3). Due to the logarithmic distribution of heat flux, subsequent deviations from the CBE change logarithmically in magnitude.

3.2. Step 2: Conductive Layer Thermal Properties

Prior to calculating Europa's icy layer thicknesses, it is first necessary to determine the thermal properties of the conductive layer, which are functions of temperature, porosity, and composition (Equation (7)). To place bounds on the temperature differential across the conductive layer, a mean surface temperature distribution is first derived (Table 3). The temperature at the base of the conductive layer depends on the behavior of the convective layer.

Convection is an inherently multidimensional process, and significant efforts have been made to suitably parameterize this process in one dimension (e.g., Moresi & Solomatov 1995; McKinnon 1999). The viscosity of Europa's icy shell is exponentially dependent on temperature, and varies many orders of magnitude from the cold surface to the warm base. This large viscosity variation places Europa within the stagnant lid regime (e.g., McKinnon 1999; Howell & Pappalardo 2019),

Table 2
List of Input Parameters, Their Corresponding Range of Values, Uncertainties, and Assumed Distributions with Corresponding Certainties

Description	Symbol	Range	Units	Cert.	Distro.	References
Mean surface temperature	T_{surf}	104.0 ± 7.0	K	1σ	normal	Ojakangas & Stevenson (1989), Spencer et al. (1999), Ashkenazy (2019)
Porosity curing temperature	T_{ϕ}	150 ± 20	K	3σ	normal	Nimmo et al. (2003), Johnson et al. (2017), Howell & Pappalardo (2019)
H ₂ O layer thickness	$D_{\text{H}_2\text{O}}$	127 ± 21	km	1σ	normal	Anderson et al. (1998), Kuskov & Kronrod (2005), Casajus et al. (2020)
Iron core radius	D_{iron}	630 ± 180	km	3σ	normal	Anderson et al. (1998), Tobie et al. (2003), Kuskov & Kronrod (2005)
Radiogenic specific heat production	H_{rad}	$(4.5 \pm 1.0) \times 10^{-12}$	W kg ⁻¹	1σ	normal	Spohn & Schubert (2003), Hussmann et al. (2010)
Icy shell rigidity	$\mu_{E,\text{ice}}$	3.5–0.5	GPa	1σ	truncated normal	Gammon et al. (1983), Nimmo & Schenk (2006), Sinha (1989)
Upper icy shell porosity	ϕ	0.0–0.3	...	...	uniform	e.g., Nimmo et al. (2003)
Upper icy shell salt fraction	f_s	$3 \times 10^{-2} \pm 1$ order	...	3σ	lognormal	e.g., Hand & Chyba (2007)
Salt reference density	ρ_s	2086 ± 501	kg m ⁻³	3σ	lognormal	Average and standard deviation of NaCl (2160 kg m ⁻³), MgSO ₄ (2660 kg m ⁻³), and natron (1440 kg m ⁻³)
Salt thermal conductivity coefficient	B_k	10^{-1} – 10^1	...	...	loguniform	Urquhart & Bauer (2015)
Global mean tidal flexing strain amplitude	ϵ_0	$(1.0 \pm 0.05) \times 10^{-5}$	...	1σ	normal	Showman & Han (2004), Nimmo et al. (2007)
Ice grain size	d	$1 \times 10^{-3} \pm 1$ order	m	1σ	lognormal	De La Chapelle et al. (1998), Barnes et al. (2002), Barr & McKinnon (2007)
Ice grain boundary width	δ	$(7.1 \pm 1.9) \times 10^{-10}$	m	1σ	normal	Frost & Ashby (1982), Hondoh (2019)
Volume diffusion prefactor	$D_{0,v}$	$(9.1 \pm 0.3) \times 10^{-4}$	m ² s ⁻¹	1σ	normal	Ramseier (1967), Goldsby & Kohlstedt (2001)
Grain boundary diffusion prefactor	$D_{0,b}$	$(8.4 \pm 0.3) \times 10^{-4}$	m ² s ⁻¹	1σ	normal	Ramseier (1967), Goldsby & Kohlstedt (2001)
Volume diffusion activation energy	Q_v	$(59.4 \pm 3.0) \times 10^3$	J mol ⁻¹	1σ	normal	Ramseier (1967), Goldsby & Kohlstedt (2001)
Grain boundary diffusion activation energy	Q_b	$(49.0 \pm 2.4) \times 10^3$	J mol ⁻¹	1σ	normal	Goldsby & Kohlstedt (2001)

Table 3
Model Predictions for Current Best Estimate (CBE) Values and 1σ Uncertainties Extracted from Derived Distributions

Description	Symbol	CBE	Units
Total icy shell thickness	D_{ice}	$24.3^{+22.8}_{-1.5}$	km
Conductive layer thickness	D_{cond}	$10.4^{+5.8}_{-1.9}$	km
Convective layer thickness	D_{conv}	$5.8^{+6.3}_{-0.8}$	km
Porous ($\sim$ brittle) layer thickness	D_{phi}	$3.2^{+4.2}_{-0.7}$	km
Example observationally constrained total thickness ^a	$D_{\text{ice,constrained}}$	$24.2^{+12.5}_{-1.4}$	km
$\log_{10}(\text{Convective ice viscosity})$	$\log_{10}(\eta_{\text{ice,conv}})$	$14.7^{+0.3}_{-0.4}$	$\log_{10}(\text{Pa s})$
Surface heat flux ^b	$\dot{q}_{\text{surf}}$	$24.5^{+3.9}_{-10.8}$	mW m^{-2}
Total rocky heat flux ^b	$\dot{q}_{\text{rock,total}}$	$10.5^{+1.7}_{-2.1}$	mW m^{-2}
Radiogenic heat flux ^b	$\dot{q}_{\text{rock,rad}}$	$7.4^{+0.7}_{-0.2}$	mW m^{-2}
Silicate tidal flux ^b	$\dot{q}_{\text{rock,tidal}}$	$3.5^{+0.6}_{-0.1}$	mW m^{-2}
Ice tidal heat flux ^b	$\dot{q}_{\text{ice,tidal}}$	$30.9^{+24.2}_{-2.6}$	mW m^{-2}
Conductive layer thermal conductivity	$\bar{k}(T, \phi, f_3)$	$3.2^{+0.2}_{-0.1}$	$\text{W m}^{-2} \text{K}^{-1}$
Porous fraction of conductive layer	f_{ϕ}	$0.43^{+0.03}_{-0.03}$	...
Isothermal temperature of convection	T_{conv}	$251.6^{+0.37}_{-1.1}$	K
Conductive layer density	ρ_{cond}	925^{+16}_{-27}	kg m^{-3}
Convective layer density	ρ_{conv}	$918^{+0.1}_{-0.3}$	kg m^{-3}
Moment of Inertia Factor	MOI	$0.359^{+0.003}_{-0.003}$	...

Notes. Note that the values listed are conditioned by the distribution of D_{ice} because the variables are not truly independent; they are optimized as a whole, rather than individually, to produce full icy shells. Thus, each value and range listed below is influenced by each of the other values and ranges, because each of these informs the distribution of D_{ice} . For example, the CBE layer thicknesses do not sum to the CBE total thickness, because models of full icy shells that exhibit both the CBE conductive and convective thickness are rare. See Section 4.2 for the full discussion.

^a In Section 5.2, I overview how future observations and predictions can help constrain the results produced by the model described in this paper. This is demonstrated by example, treating the inferred and predicted icy shell and layer thickness estimates of Billings & Kattenhorn (2005) as independent observations.

^b The upper and lower bounds for heat fluxes are nonlinear. The distributions themselves are logarithmic, and so I report the values that represent only the first standard deviation from the mean, noting that subsequent deviations change logarithmically in magnitude.

where an conductive upper lid forms and does not participate in convective heat transfer or flow.

The isothermal temperature of convection is related to the vigor of convective flow, which depends on both the thickness of a layer as well as its internal heat generation (e.g., Moresi & Solomatov 1995). Because these values are interdependent, they cannot be applied in the Monte Carlo approach without requiring computationally prohibitive iterative solutions. Instead, I make the following assumptions to create a self-consistent approximation for the isothermal temperature of convection, T_{conv} .

The viscosity ratio across the icy shell, $\Delta\eta$, is related to T_{conv} by (McKinnon 1999)

$$\ln(\Delta\eta) = \frac{Q_v}{R} \frac{T_{\text{melt}} - T_{\text{surf}}}{T_{\text{conv}}^2} = \Theta, \quad (8)$$

where R is the universal gas constant, Q_v is the volume diffusion activation energy (Table 2) because volume diffusion terms dominate the viscosity solution in the region of interest, and Θ is a measure of e -foldings (the number of times the value of a variable, in this case η , increases by a factor of e).

According to Moresi & Solomatov (1995), the temperature difference between the base of the convecting layer and the convection cell is of the order

$$T_{\text{melt}} - T_{\text{conv}} \sim \frac{T_{\text{melt}} - T_{\text{surf}}}{\Theta}. \quad (9)$$

Assuming that the relationship in Equation (9) provides an approximate equality, then Θ can be written as a nondimensional temperature ratio related to the efficiency of convection in the layer (e.g., Turcotte & Schubert 2014, chap. 6.19–6.21):

$$\Theta = \frac{T_{\text{melt}} - T_{\text{surf}}}{T_{\text{melt}} - T_{\text{conv}}}. \quad (10)$$

An approximation of T_{conv} can now be made from Equations (8)–(10):

$$T_{\text{conv}} \approx \frac{\sqrt{4T_{\text{melt}}R/Q_v + 1} - 1}{2R/Q_v}. \quad (11)$$

Because the temperature profile associated with vigorous convection is approximately symmetrical with depth, it is then possible to estimate the temperature at the base of the conductive portion of the icy shell:

$$T_{\text{cond,base}} = 2T_{\text{conv}} - T_{\text{melt}}. \quad (12)$$

With the temperatures of critical interfaces within the ice shell now estimated, I can calculate the remaining thermal properties. The temperature dependence of thermal conductivity for pure ice I is approximated according to Nimmo & Manga (2008):

$$k(T) = 567/T [\text{K}]. \quad (13)$$

This inverse temperature dependence results in a steady-state temperature as a function of depth, z , within an icy shell of (e.g., Johnson et al. 2017)

$$T(z) = T_{\text{surf}}(T_D/T_{\text{surf}})^{z/D}. \quad (14)$$

Following Equations (13) and (14), the temperature-dependent thermal conductivity averaged over a depth below the surface, D , to a temperature of $T(z) = T_D$, is given by

$$\begin{aligned} \bar{k}(T) &= \frac{1}{D} \int_0^D \frac{567[1/\text{K}]}{T_{\text{surf}}} (T_D/T_{\text{surf}})^{z/D} dz \\ &= \frac{567[1/\text{K}](T_{\text{surf}} - T_D)}{\ln(T_D/T_{\text{surf}})T_{\text{surf}}T_D}. \end{aligned} \quad (15)$$

To implement Equation (15), it is necessary to consider that the conductive portion of the icy shell is not uniform with depth. Therefore, the model next considers the effect of porosity in the cold, upper portion of the conductive layer. Primarily due to fracturing, the porosity of the icy lithosphere of Europa may vary by tens of percent across the satellite (e.g., Nimmo et al. 2003), affecting both the density and thermal conductivity (Table 3).

The lithosphere porosity will extend to a depth where the overburden pressure is large enough and the viscosity is low enough to promote viscous relaxation of the pore space (e.g.,

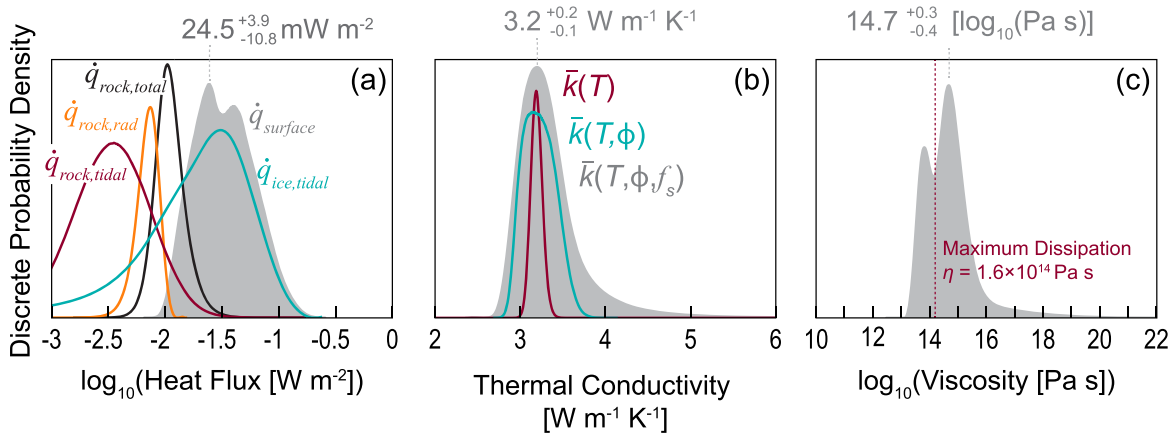


Figure 3. Derived discrete probability distributions with most likely values labeled as current best estimates (Table 3). (a) Distribution in heat flux for the surface (gray), showing contributions from icy tidal heating (blue) and the rocky interior (black). Also shown are the silicate tidal heat flux (red), and heating from radioactive decay (orange). (b) Distribution in conductive layer thermal conductivity (gray), showing the temperature dependence (red) and contribution of the porous layer (blue). (c) Distribution in interior convective viscosity.

Nimmo et al. 2003; Johnson et al. 2017; Howell & Pappalardo 2019) over the 40–90 Myr surface age of the icy shell (Bierhaus et al. 2009). This porosity curing temperature, T_ϕ , depends on the age of porosity emplacement, the thermal history of the icy shell, and the specific processes that introduce porosity. A distribution for T_ϕ is obtained by sampling values from the literature (Table 2). A uniformly sampled porosity of 0%–30% is applied to the depth, z , where $T(z) = T_\phi$. Following Equation (15), the temperature- and porosity-dependent depth-averaged thermal conductivity can be calculated:

$$\bar{k}(T, \phi) = (1 - f_\phi)\bar{k}(T) + f_\phi(1 - \phi)\bar{k}(T). \quad (16)$$

Here, f_ϕ is the fraction of the conductive layer thickness that is porous. The porous fraction can be found from the temperature profile through change of variables

$$f_\phi = \frac{\ln(T_\phi/T_{\text{surf}})}{\ln(T_{\text{cond,base}}/T_{\text{surf}})}. \quad (17)$$

Finally, non-ice contributions are added to the conductive layer. The bulk abundance of non-ice materials trapped within the icy shell increases with increasing freezing rate, and is limited to the abundance of non-water materials within the ocean (e.g., Buffo et al. 2020). This study assumes that salts are only contained in significant quantities within the conductive region of the icy shell because the freezing rates below the conductive portion of the icy shell are likely too low to provide significant non-ice material entrainment (Buffo et al. 2020).

To bound the unknown salinity of Europa’s icy shell, I invoke estimates of Europa’s ocean salinity of $\sim 3\%$ (e.g., Hand & Chyba 2007), approximately an order of magnitude less than the eutectic abundance of salts within the icy shell. Thus, it is justified to assume a log-normally distributed f_s with mean 3×10^{-2} and deviation of $3\sigma = 1$ order of magnitude to maintain physical bounds.

Capturing the effect of non-ice constituents on thermal conductivity is not straightforward. Critical supporting laboratory data are sparse (e.g., Prieto-Ballesteros & Kargel 2005), lacking in surveys of the thermal properties of salt–ice mixtures formed under various freezing conditions and across a range of relevant temperatures and pressures. When excluded to the ice grain boundaries, salts like NaCl may enhance the overall thermal conductivity significantly (Cox & Weeks 1975;

Urquhart & Bauer 2015), while hydrated salts (Kumar et al. 2019) and gas hydrates (Kalousova & Sotin 2020) may be insulating.

Due to the lack of sufficient data, this study assumes that it is equally likely salts and other non-ice constituents degrade or enhance thermal conductivity. This is introduced in the form of a scaling coefficient of the effective conductivity within the icy shell, B_k , which is sampled from a log-uniform distribution spanning 10^{-1} – 10^1 . The resulting thermal conductivity of non-ice materials spans an order of magnitude above and below the pure ice thermal conductivity:

$$\bar{k}(T, \phi, f_s) = (1 - f_s)\bar{k}(T, \phi) + f_s B_k \bar{k}(T, \phi). \quad (18)$$

A range of various salt species relevant to Europa are used to determine the salt density distribution (Table 2), including NaCl and MgSO_4 (Zolotov & Shock 2001; Hand & Chyba 2007), and natron (McCord et al. 1999; Johnson et al. 2017). While this study assumes mineral salt densities are consistent with previous icy shell studies (e.g., Johnson et al. 2017; Howell & Pappalardo 2019), these salts may also form hydrates that affect their densities in a manner that is not modeled here. Melting temperatures at the base of the icy shell are calculated from a lookup table assuming NaCl. The final distributions for thermal conductivity are shown in Figure 3(b). The CBE thermal conductivity of the conductive layer is $3.2^{+0.2}_{-0.1} \text{ W m}^{-1} \text{ K}^{-1}$ (Table 3).

3.3. Step 3: Bound Conductive Thickness

After obtaining a heat flux contribution for the rocky interior of Europa, the thermal properties of the conductive icy shell layer, and the temperatures at the various icy shell interfaces, I now bound the distribution for conductive layer thickness. As shown in Equation (7), the conductive thickness is inversely proportional to the flux of heat escaping the body.

A maximum bound on the conductive thickness for a given Monte Carlo iteration can therefore be obtained by investigating the minimum potential total heat flux. This is calculated as the heat flux from the silicate interior added to the tidal heat generated within the lower portion of a purely conductive icy shell, ignoring any minor contribution of heat generated within the ocean (Hay & Matsuyama 2019; Equation (1)). The minimum contribution to the heat flux budget of the icy shell is

estimated by replacing the convective layer with a conductive layer spanning $T_{\text{cond,base}}$ to T_{melt} , assumed for simplicity to have a bulk viscosity equal to the characteristic viscosity of the equivalent convecting layer.

Iterations with values of D_{cond} that exceed the critical Rayleigh number ($\text{Ra}_{\text{crit}} \approx 10^6$; McKinnon 1999) are rejected as nonphysical because the conductive layer would tend to convect. I assume that the uncertainty associated with the critical Rayleigh number is less than the broader rheological uncertainties associated with convection discussed below (e.g., grain size), and this uncertainty is therefore absorbed by the rheological estimations.

The maximum conductive thickness is now bounded by both conductive heat flow arguments (Equation (7)) and by the critical Rayleigh Number. The physical upper limit on the conductive icy thickness will be given by the lesser of these two values because exceeding either condition results in nonphysical solutions:

$$D_{\text{cond,max}} = \min \left\{ \bar{k}(T, \phi, f_s) \frac{T_{\text{cond,base}} - T_{\text{surf}}}{\dot{q}_{\text{rock,tidal}} + \dot{q}_{\text{rock,rad}} + \min(\dot{q}_{\text{conv,ice}})}, \left(\frac{\text{Ra}_{\text{crit}} K \eta_{\text{ice,conv}}}{\alpha(T_{\text{cond,base}} - T_{\text{surf}}) \rho_0 g} \right)^{\frac{1}{3}} \right\}, \quad (19)$$

$$K = \bar{k}/(T, \phi, f_s) \rho(T, \phi, f_s) C_p. \quad (20)$$

Here, K is the thermal diffusivity, α is the thermal expansivity, $\rho(T, \phi, f_s)$ is the density averaged over the conductive layer, C_p is the specific heat capacity of ice, ρ_0 is the reference density, and g is the gravitational acceleration.

Conversely, bounding the minimum potential conductive thickness requires an estimate of the maximum potential total heat flux through the conductive portion of the icy shell. This additional heat is provided by tidal dissipation within the convecting icy layer. The tidal heat produced within the convecting ice is proportional to the thickness of the convecting layer and is viscosity dependent (Equation (2)). Estimating a value for the maximum thickness of the convecting layer is straightforward, as the convecting layer cannot exceed the thickness of the full H_2O layer that includes the ocean and icy shell, providing $D_{\text{H}_2\text{O}}$ as a reasonable maximum bound. This assumption introduces a small number of solutions that have total icy shell thicknesses greater than the H_2O layer thickness, which are discarded as nonphysical.

In order to calculate the tidal heat generation within a convecting layer of a given thickness, it is first necessary to determine the layer viscosity (Equation (2)). For the convecting icy layer, viscosity is given by the Newtonian diffusion creep relationship for ice I (Goldsby & Kohlstedt 2001), which is the predicted deformation mechanism in the convecting interior (e.g., Howell & Pappalardo 2018):

$$\eta_{\text{ice,conv}} = \frac{RT_{\text{conv}} d^2}{42 V_m} \left(B_{0,v} e^{-Q_v/RT_{\text{conv}}} + \frac{\pi \delta}{d} B_{0,b} e^{-Q_b/RT_{\text{conv}}} \right)^{-1}. \quad (21)$$

Here, R is the universal gas constant, T is the temperature, d is the characteristic grain size, V_m is the molar volume, B_0 and Q

are respectively the preexponential term and activation energy for volume diffusion (v) and grain boundary diffusion (b), and δ is the grain boundary width, assumed here to be twice the Burgers Vector length (a quantification of the spatial scale of lattice deformation in response to a lattice dislocation; Frost & Ashby 1982; Goldsby & Kohlstedt 2001). Distributions that capture the uncertainties in the experimentally determined parameters are summarized in Table 2.

The distribution in grain size is challenging to estimate. Grain size evolves with temperature, stress and strain environment, deformational history, and local composition (e.g., Barr & McKinnon 2007). Barr & McKinnon (2007) numerically modeled the two-dimensional development of European ice convection cells with a self-consistent grain size, and found that uncertainties in the physical and compositional state of Europa's icy shell result in potential grain sizes from 10^{-3} to 10^{-1} m. They note that these findings are consistent with terrestrial measurements of icy grain sizes spanning 0.6–4 mm across a variety of geographic locations icy ages (e.g., De La Chapelle et al. 1998; Barnes et al. 2002). The temperature, pressure, and potentially the strain rates experienced by terrestrial glaciers are similar to those of the convective region of Europa's icy shell (e.g., Barr & McKinnon 2007). Given that these best estimates for Europa grain size are on the order of millimeters, and that the uncertainty lies in the order of magnitude of grain size, I conservatively sample from a log-normal distribution of potential ice grain size with mean 10^{-3} m and deviation of 3 orders of magnitude at the 3σ certainty level to capture the potential effects of the poorly constrained European ice environment.

From the distributions in T_{conv} , $B_{0,v}$, $B_{0,b}$, Q_v , Q_b , δ , and d , I am now able to derive the distribution for $\eta_{\text{ice,conv}}$ (Figure 3(c)) using Equation (21). The CBE viscosity of the convecting ice is $10^{14.7 \pm 0.3}$ Pa s (Table 3).

To derive a maximum icy tidal heat flux using Equation (2), a value for the ice rigidity, $\mu_{E,\text{ice}}$, is needed. The laboratory value for $\mu_{E,\text{ice}}$ is 3.5 GPa (Gammon et al. 1983). Studies of the elastic behavior of European ice infer relatively low Young's moduli of ~ 1 –5 GPa within the brittle portion of the icy shell (Nimmo et al. 2003; Nimmo 2004; Nimmo & Schenk 2006), corresponding to a modulus of rigidity as low as ~ 0.3 –2 GPa. A reduction in rigidity by order of magnitude below the laboratory value is likely the lower physical limit for ice on Europa (Nimmo & Schenk 2006). Therefore, I sample $\mu_{E,\text{ice}}$ from a truncated normal distribution with a maximum value of 3.5 GPa and 1σ uncertainty of 0.5 GPa (Table 2).

From Equations (1)–(7), the minimum bound on the conductive thickness can now be found for a given Monte Carlo distribution:

$$D_{\text{cond,min}} = \bar{k}(T, \phi, f_s) \times \frac{T_{\text{cond,base}} - T_{\text{surf}}}{\dot{q}_{\text{rock,tidal}} + \dot{q}_{\text{rock,rad}} + D_{\text{H}_2\text{O}} \frac{\eta_{\text{ice,conv}} \epsilon_0^2 \omega^2}{1 + \eta_{\text{ice,conv}}^2 \omega^2 / \mu_{E,\text{ice}}^2}}. \quad (22)$$

For each individual iteration, the conductive thickness now has finite bounds.

3.4. Step 4: Determine Layer and Total Thicknesses

The next step toward determining icy shell thickness is to sample the conductive layer thickness. At this stage, all

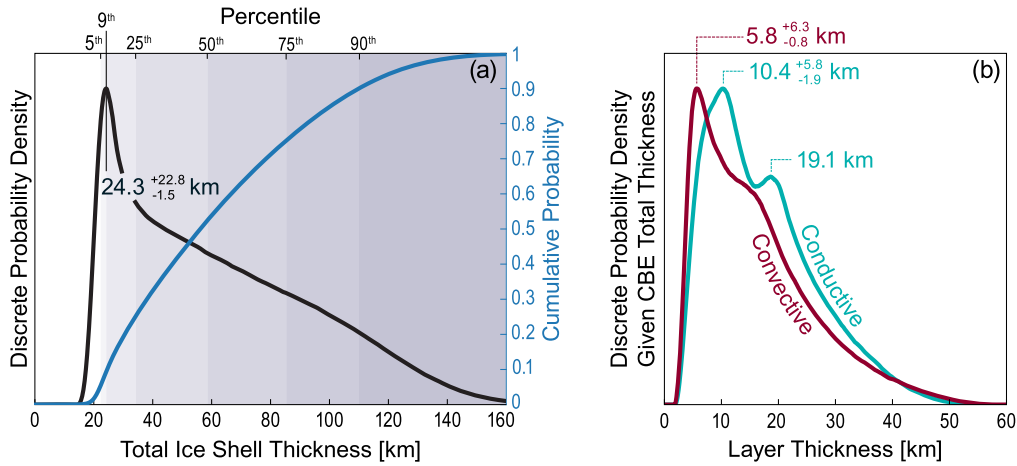


Figure 4. Icy thickness probability distributions predicted by the Monte Carlo model. (a) The highly skewed discrete probability density distribution (black line) for total icy shell thickness, and cumulative probability distribution (blue line). Percentiles are labeled above, with only a $\sim 9\%$ probability of the icy shell being as thin or thinner than the CBE value of $24.3^{+22.8}_{-1.5}$ km. (b) Conditional discrete probability distributions for convective and conductive thickness, given the CBE total shell thickness, $P(D_{\text{conv}}|D_{\text{ice,CBE}})$ and $P(D_{\text{cond}}|D_{\text{ice,CBE}})$. Note that no one specific convective and conductive thickness sum to create the CBE total icy shell; rather, a spectrum of variable layer thicknesses are represented by the peak and tails presented in (a).

independent information related to the overarching heat flux balance (Equation (1)) has been introduced in the appropriate form through the selection of the various input distributions. Because there is no additional independent information on which to base a choice of distributions in conductive layer thickness, I sample conductive thickness uniformly between the derived bounds.¹

After sampling a value for D_{cond} , the implied value of D_{conv} can be uniquely calculated from Equations (1)–(7), as each iteration has now sampled all relevant values:

$$D_{\text{conv}} = \frac{1 + \eta_{\text{ice,conv}}^2 \omega^2 / \mu_{E,\text{ice}}^2}{\eta_{\text{ice,conv}} \epsilon_0^2 / \omega^2} \left[\bar{k}(T, \phi, f_s) \times \frac{T_{\text{cond,base}} - T_{\text{surf}}}{D_{\text{cond}}} - (\dot{q}_{\text{rock,tidal}} + \dot{q}_{\text{rock,rad}}) \right]. \quad (23)$$

The total icy shell thickness is then simply calculated as $D_{\text{ice}} = D_{\text{cond}} + D_{\text{conv}}$. The resulting distributions in the layer and total thicknesses are shown in Figures 4 and 5.

4. Results and Discussion

4.1. Input Parameter Sensitivities

The Monte Carlo model employed in this study ingests both accepted standard values for Europa (Table 1) and quantities with significant theoretical or observational uncertainties (Table 2). To discover the largest driving uncertainties in our understanding of Europa’s icy thickness, I perform a Spearman’s rank correlation test between the input parameters with significant uncertainties and the layer and icy shell thickness predictions (Table 4).

As shown above, these underlying distributions and their mathematical relationships are often nonlinear, rendering a linear correlation like the more prevalent Pearson moment product inappropriate for this application (e.g., Hauke & Kossowski 2011).

¹ An alternate choice for a distribution of D_{cond} might be to employ a power law governed by the inverse relationship between conductive thickness and heat flux (Equation (7)). More energy is required to produce thinner icy shells, and therefore thinner shells are less likely to manifest. However, that inverse relationship is already present within the model, and thus such a selection would likely introduce or compound bias.

Spearman’s rank correlation was chosen because it tests for any monotonic relationship between variables (e.g., Hauke & Kossowski 2011). The correlation coefficient tests the hypothesis that for a given increase in variable x , there is always an increase or always a decrease in variable y . If an increase in x always produces an increase in y , the correlation coefficient is 1. If an increase in x always produces a decrease in y , the correlation coefficient is -1 . Two variables are uncorrelated if an increase in x is equally likely to result in either an increase or decrease in y , in which case the correlation coefficient is 0.

In addition to determining the correlation coefficient, I also test for the significance of the correlation, reported as a p -value (e.g., Hauke & Kossowski 2011). This value characterizes the probability that any apparent correlation is simply the result of random chance due to biased sampling of an uncorrelated population. A 95th percentile confidence level in the correlation corresponds to a p -value ≤ 0.05 .

The prominent controls on conductive layer thickness include the tidally generated heat flux from the icy shell, driven by ice viscosity and the relevant dependencies, heat generated within the rocky interior, and the conductive layer thermal conductivity, which is sensitive to the layer porosity and the effects of salts. Convective layer thickness is primarily sensitive to the layer viscosity, which controls heat generation in the convecting layer per unit depth. The convective thickness is secondarily sensitive to the total H_2O layer thickness that provides its upper bound. The total icy shell thickness is most sensitive to ice rheology because tidal dissipation in the convective layer is the majority contributor to the total heat flux (Figure 4).

4.2. Derived Distributions

In order to produce predictions for icy shell thickness, several underlying distributions were derived. As introduced in the antecedent section, three derived distributions of primary interest are reported in Figure 3, and the remaining distributions are summarized in Table 3.

Tidal heat generation within Europa’s icy shell plays a crucial role in the evolution and state of the shell and of the

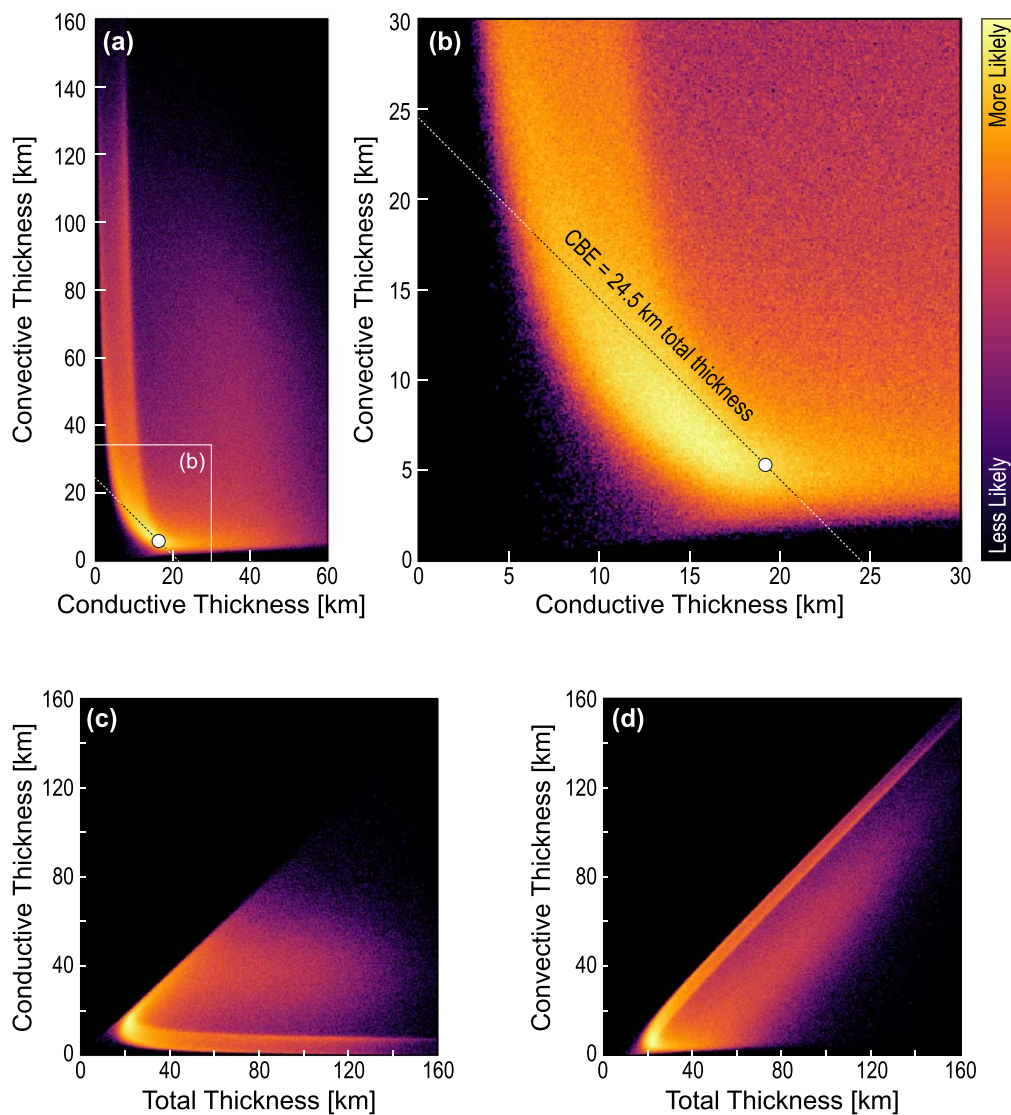


Figure 5. Two-dimensional joint-probability density distributions for icy shell and layer thicknesses. (a) Joint-probability distribution for layer thicknesses. Lines of constant icy shell thickness can be drawn between identical values on the horizontal and vertical axes. The white box highlights the location of inset (b). In (a), (b), the current best estimate (CBE) total thickness (dashed line) is related to probability distributions for layer thicknesses reported in Figure 4(b). The CBE icy shell is marked with a white circle. This CBE value is approximately consistent with the CBE value for convective layer thickness, and with the less-favored peak in the bimodal distribution for conductive layer thickness (Figure 4(a)). In (c) and (d), conditional probability distributions show how the total icy shell thickness varies with conductive and convective thickness.

body as a whole. The primary contributor to the global heat flux balance is tidal dissipation within the convective icy layer. A minority contribution is provided by radioisotope decay in the silicate interior, with a relatively negligible contribution coming from silicate tidal dissipation. The peak probability and thus current best estimate value for total surface heat flux is $24.5^{+3.9}_{-10.8}$ mW m⁻².

The thermal conductivity of the conductive portion of the icy shell influences numerous tectonic and geodynamic processes. Over the relevant temperature range, the temperature dependence of thermal conductivity alone produces a well-constrained probability distribution. Porosity within the upper portion of the conductive layer lowers the thermal conductivity because heat is transferred inefficiently across the vacuum-filled pores. This results in a broadening of the tails of the distribution and a slight skew toward lower values. For simplicity and because of the sparsity of data, non-ice contributions are assumed to be equally likely to significantly enhance thermal conductivity as

to significantly diminish thermal conductivity. Because porosity has a strong diminishing effect, the additional contribution of non-ice material primarily lengthens the tail of the distribution rightward. The resulting distribution is slightly skewed to favor higher conductivity values, with a CBE value of $3.2^{+0.2}_{-0.1}$ W m⁻¹ K⁻¹.

The viscosity of the convecting icy interior is key to descriptions of convection within Europa's icy shell (Mitri & Showman 2005; Barr & McKinnon 2007; Barr & Showman 2009), the tidal response of the icy shell (Tobie et al. 2003; Barr & Showman 2009; Sotin et al. 2009), and the stress state of the shell as it deforms and modifies the surface (e.g., Howell & Pappalardo 2019). Canonically, studies of Europa's icy shell assume values of 10^{13} – 10^{15} Pa s, one order of magnitude above and below the viscosity at which tidal heat production is maximized ($\sim 1.5 \times 10^{14}$ Pa s) (e.g., Mitri & Showman 2005; Sotin et al. 2009; Howell & Pappalardo 2018).

Table 4

Characterization of Spearman's Rank Correlation Coefficients between Icy Layer Thicknesses, and Input Parameters and Derived Distributions

Description	Symbol	Correlation Coefficient (<i>c</i>)		
		D_{ice}	D_{cond}	D_{conv}
H ₂ O layer thickness	D_{H_2O}	0.141	0.004	0.140
Surface heat flux	$\dot{q}_{surface}$	-0.306	-0.984	0.199
Total rocky heat flux	$\dot{q}_{rock,total}$	-0.087	-0.165	0.026
Radiogenic heat flux	$\dot{q}_{rock,rad}$	-0.039	-0.044	0.001
Silicate tidal heat flux	$\dot{q}_{rock,tidal}$	-0.074	-0.146	0.023
Ice tidal heat flux	$\dot{q}_{ice,tidal}$	-0.291	-0.927	0.172
Convective ice viscosity	$\eta_{ice,conv}$	0.472	0.828	0.086
Grain size	d	0.439	0.752	0.091
Volume diffusion activation energy ^a	Q_v	0.093	0.167	0.012
Conductive layer thermal conductivity	$\bar{k}(T, \phi, f_s)$	0.068	0.105	-0.006
Porosity	ϕ	-0.030	-0.044	0.006
Salt fraction	f_s	0.023	0.034	0.003
Salt thermal conductivity coefficient	B_k	0.042	0.066	-0.005

Notes. These correlations test for any monotonic relationship. Larger absolute values correspond to stronger correlations. Of ~40 tested relationships, this table highlights those with strong correlations ($|c| \geq 0.1$) and relevant dependencies, to add context. All correlations shown are significant at the 95% confidence level. See Section 4.1.

^a Grain boundary diffusion activation energy exhibited no significant correlations.

This study predicts a CBE viscosity for the convecting layer of $10^{14.7 \pm 0.3}$ Pa s, with the canonical value falling more than one full standard deviation outside the CBE. The viscosity distribution appears bimodal because it is sensitive to icy shells dominated both by conduction and by convection (Figure 5). Intriguingly, there are a paucity of solutions at the viscosity that maximizes tidal heat production when considering joint distributions informed by the full range of model parameters. Viscosities below 10^{13} Pa s and above 10^{18} Pa s, where the importance of tidal heating is diminished compared to radiogenic heating (Sotin et al. 2009), are also unlikely in the warmer, ductile region of the icy shell. Uncertainty in the convective ice viscosity is primarily driven by the poorly characterized range of potential grain sizes and errors in experimental constants contained within the viscosity exponent (Equation (21); Table 2).

4.3. Icy Shell and Layer Thicknesses

To understand Europa's full icy shell thickness D_{ice} , I first consider the full joint-probability distribution for conductive and convective thicknesses, $P(D_{conv} \cap D_{cond})$ (Figures 5(a) and (b)). The resulting joint-probability distribution is bimodal (a “fishhook” shape in two dimensions) from ~10 to 40 km total thickness. This is pronounced in joint-probability distributions of the full thickness and each of the individual layers, which display small secondary tails of high likelihood (Figures 5(c) and (d)).

The bimodal distributions in layer thickness correspond to icy shell thicknesses that have two preferred solutions, one dominated by conduction and one dominated by convection. For the same total thickness, there exists a solution with a thick

conductive lid and low surface heat flux, and a solution with a thin conductive lid supported by additional heat production in a thick convecting layer. For icy shells thicker than ~40 km, convection dominates, while conduction dominates in thin shells (Figure 5). Conductive layers are almost always the minority contributor to icy shells thicker than ~40 km.

The model predicts a distribution for icy shell thickness that is highly skewed, with a CBE of $24.3^{+22.8}_{-1.5}$ km, spanning the full range of potential hydrosphere thicknesses (Figure 4(a)). A model result one standard deviation above the CBE is ~100% thicker, while a result one standard deviation below the CBE is only ~6% thinner. The steep slope of the probability distribution at low thicknesses is a result of the inverse relationship between conductive thickness and heat flux (Equation (7)): as the conductive thickness approaches zero, the heat flux through the layer must approach infinity. The shallow slope in the probability distribution at large icy shell thicknesses is primarily a result of the heat flux from the interior and convective layer of the icy shell. Convective layers span a wide range in tidal energy production per unit depth, and lower interior heat fluxes require larger convective layers to produce the same total heat flux through the icy shell.

Evaluated at the CBE thickness by convolving the full solution with the distribution for D_{ice} , the distributions for the thicknesses of the individual conductive and convective layers are shown in Figure 4(b). The distributions are heavily skewed and bimodal, with a CBE conductive thickness of $10.4^{+5.8}_{-1.9}$ km, and CBE convective thickness of $5.8^{+6.3}_{-0.8}$ km.

It is important to note that the CBE values for the icy layer thicknesses do not sum to the CBE total shell thickness. For example, an ice shell near the 24.3 km CBE total thickness might have a convective layer thickness near the corresponding 5.8 km CBE value, but a conductive layer thickness nearer to the less-favored peak of the bimodal distribution of 19.1 km (Figure 4). The full icy shell thickness is influenced by complex relationships between the underlying distributions, and the models that produce total thicknesses near the CBE value rarely also predict both the conductive and convective layers to be near their respective CBE value. Indeed, when optimizing across all variables, and when enforcing the overarching equations that maintain physical relationships, the most common values for one variable are often paired with less likely values for another. As with the icy shell thicknesses, it should be expected that the CBE total thickness does not necessarily correspond to the CBE values for all of the parameters summarized in Table 3.

5. Perspectives

5.1. A Dynamic Icy Shell

The steady-state model described here may provide a path for estimating the dynamic state of Europa's icy shell. For example, consider that a future observation of Europa confirms the icy shell to be 5 km thick. From this work, one might interpret that (1) the icy shell is most likely dominated by conduction because the Rayleigh number, which depends on D^3 , would be unlikely to exceed the critical value within such a thin shell (e.g., McKinnon 1999), and (2) the icy shell thickness is many standard deviations below the predicted CBE. Such a low-probability measurement could imply that the anomalous heat flux is related to the release of the latent heat of fusion of ice at the base of the shell freezing as the shell thickens through

time. The thickening rate of such a shell can be estimated by investigating the heat flux anomaly required to achieve the observed conductive thickness:

$$\frac{dD}{dt} = \frac{1}{\rho L} \left(\frac{\bar{k}(T, \phi, f_s)(T_{\text{cond,base}} - T_{\text{surf}})}{D_{\text{cond}}} - (\dot{q}_{\text{rock,tidal}} + \dot{q}_{\text{rock,rad}}) \right), \quad (24)$$

where D_{cond} is the measured value of 5 km, and L is the latent heat of fusion of water (334 kJ kg⁻¹; Spencer et al. 1999). Substituting the CBE values for the remaining terms (Table 3), a 5 km icy shell could imply an icy shell thickening rate of $\sim 9 \text{ km Myr}^{-1}$ corresponding to an 84 mW m^{-2} heat flux anomaly.

Alternatively, such measurements of anomalous icy shell thickness might indicate that the silicate interior and/or ocean are generating heat at much higher or lower rates than expected, challenging assumptions about tidal dissipation in those layers or potential activity in the silicate interior. For example, the total 95 mW m^{-2} heat flux required to maintain a 5 km conductive ice shell corresponds to a k_2/Q of ~ 0.005 (Equation (5)), approximately one-third that of Io (Lainey et al. 2009). Thus, maintaining and updating probabilistic models for critical interior properties will allow quantitative investigation of any potentially unlikely observations by future exploration.

At the opposite extreme, consider an icy shell that is confirmed by a future observation to be 100 km thick. Here, one might interpret that (1) the icy shell is most likely dominated by convection (Figure 5), and (2) the icy shell thickness is ~ 4 standard deviations above the predicted CBE. In this case, the icy shell is very likely to be dominated by convection (Figure 5), and significant heat should be tidally dissipated within the thick convecting layer. One possibility is that the negative heat flux anomaly associated with the thick shell (-5 mW m^{-2}) is associated with a heat lost to the conversion of the solid phase of ice to the liquid phase.

When both the thickness of the convective region and conductive region are unknown, it is not possible to produce a unique solution for thinning rate. An attempt can be made to bound the upper limit of the rate at which the shell might be thinning by assuming there is no tidal heat dissipation within the ice, and that the conductive thickness is equal to the full icy shell thickness. For a 100 km icy shell, an upper limit on the thinning rate would be $\sim 5 \text{ km Myr}^{-1}$. Alternatively, such a thick shell and limited heat budget may indicate that Europa accreted from material depleted in the isotopes responsible for radiogenic heating (e.g., Hussmann et al. 2010).

5.2. Incorporating Inferred Constraints

Our knowledge of Europa will continue to grow through telescopic remote observations and with the planned NASA Europa Clipper and ESA JUICE missions, and through the planned extension of the NASA Juno mission. Eventually, the interior of the icy shell may be profiled remotely by a potential future lander or in situ by a subsurface probe (Hand et al. 2017; Hendrix et al. 2019; Howell et al. 2020).

Near-term radar soundings of the icy shell by the Europa Clipper and JUICE spacecraft may provide information on layer thicknesses (e.g., Blankenship et al. 2009; Di Paolo et al. 2016; Kalousová et al. 2017), greatly improving our

understanding of the thermal and mechanical state. Potential near-term measurements of the deep interior structure are also crucial. For example, the CBE moment of inertia factor predicted by this Monte Carlo study is 0.359 ± 0.003 , consistent with recent predictions from reanalysis of Galileo tracking data of 0.355 ± 0.002 (Casajus et al. 2020). Revisions of this value by future missions will allow models to be filtered based on compliance with the moment of inertia constraint. Looking forward, a potential future landed mission could probe and confirm the interior structure of the icy shell and ocean through seismic investigation (Panning et al. 2018; Stähler et al. 2018), enabling a deeper understanding of the underlying processes that control layer and total thicknesses.

To illustrate how future constraints might be incorporated, I look to the previously published studies of full icy shell and brittle layer thicknesses summarized in Billings & Kattenhorn (2005) and treat them as independent observations. First, I survey the compiled predictions of total icy shell thickness inferred from studies of isostasy, impact cratering, thermal state, force balances, and Earth analogs. Counting the minimum and maximum predicted value of each study as a data point, I derive a mean and standard deviation for total icy shell thickness. Note that this exercise is not complete and may suffer from sampling bias, as compiling all recent thickness predictions is outside the scope of this study, and no effort was made to weight studies that rely on different methods for inferring icy shell thickness.

For studies with unbounded minima, I substitute a lower bound of 0 km. For studies with unbounded maxima, I substitute the 90th percentile thickness from this study of 110.4 km. For total icy shell thickness, the result is $42.0 \pm 46.6 \text{ km}$. I then convolve the corresponding truncated normal distribution with the theoretical distribution predicted by this study. The resulting distribution has a similar CBE, but it shows reduced probability of producing icy shells thicker than the CBE value (Figure 6(a)).

Next, I repeat the survey but for inferences of brittle thickness ($1.7 \pm 2.1 \text{ km}$). The depth to which the icy shell experiences brittle failure depends on the thermal and mechanical state of the shell, the rheology, the timescale of deformation, and the applied stress associated with each individual geologic event. For this comparison, I approximate the brittle thickness in the Monte Carlo simulation as the thickness of the porous region of the icy shell, beneath which any inactive fractures would have healed viscously over the surface age of the satellite. Applying the brittle thickness constraint shifts the probability distribution for total icy shell thickness to slightly lower values, and this slightly increases the probability corresponding to icy shells ~ 40 – 140 km thick. The increase in probability for thicker shells occurs because greater heat generation, and thus greater convective thickness, is required to maintain thinner conductive layers (Equation (7)).

Finally, I apply both the full thickness constraint and the brittle thickness constraint to the study predictions. The final result constrained by the compiled values of Billings & Kattenhorn (2005) is shown in Figure 6(b). The inferred thickness constraints primarily act to reduce the probability of thick icy shells, lowering the 1σ uncertainty for thicker values by a factor of 2.

These constraints fail to substantially shift the CBE icy shell thickness due to the diminishing effect of increased heat flux on decreased total icy shell thickness, as discussed above.

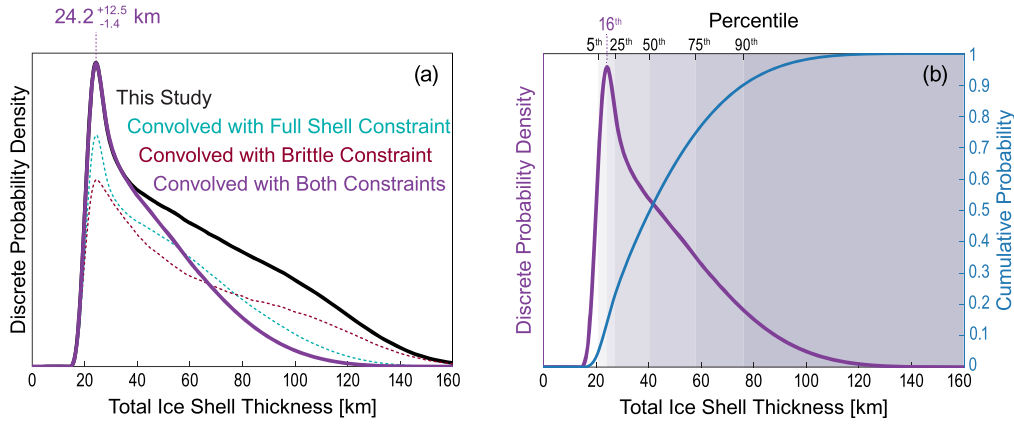


Figure 6. Icy thickness probability distributions predicted by the Monte Carlo model and filtered using studies compiled by Billings & Kattenhorn (2005). (a) Distributions for total icy shell thickness. The prediction of this study is indicated by a black line. The prediction of this study was then convolved with simple normal distributions for icy shell and brittle thicknesses compiled by Billings & Kattenhorn (2005), and the resulting convolutions are showed with colored dashed lines. (b) Same as Figure 4(a), but considering both the full thickness and brittle thickness constraints.

Effectively, the inverse relationship between conductive thickness and total heat flux acts to anchor the CBE value.

6. Conclusions

In this study, I have quantified a CBE for Europa’s icy shell thickness and the conductive and convective layers it comprises by using a multivariate Monte Carlo model to balance steady-state heat flux throughout the body. The CBE thickness for Europa’s icy shell predicted by this study is $24.3^{+22.8}_{-1.5}$ km. Icy shells $\lesssim 40$ km thick have both a predominantly convective and predominantly conductive solution. Icy shells thicker than ~ 40 km are dominated by convection. Surprisingly, only $\sim 9\%$ of models produce icy shells thinner than the CBE value of 24.3 km. Only a few percent are thinner than 15 km (Figure 4), and half of all model outcomes produce icy shells thicker than ~ 50 km. To highlight the skewness of this result, the 1σ range of icy shell thicknesses is ~ 23 –47 km, and the 2σ range is ~ 21 –70 km. Icy shells canonically regarded as very thick, e.g., 50–150 km (e.g., Carr et al. 1998; Vilella et al. 2020), are substantially more probable based on the current knowledge of the body than canonically thin shells of a few kilometers (e.g., Greenberg et al. 1999).

This study also predicts a CBE surface heat flux of $24.5^{+3.9}_{-10.8}$ mW m $^{-2}$, dominated by tidal dissipation within the icy shell, and a CBE thermal conductivity within the conductive icy layer of $3.2^{+0.2}_{-0.1}$ W m $^{-1}$ K $^{-1}$. Note again that individual CBE values are unlikely to align in models that couple distributions across many different constraints, and therefore it is inappropriate to simply add, e.g., the CBE rocky and icy heat fluxes to achieve the CBE surface heat flux.

The derived distribution for convective layer viscosity results in a CBE value of $10^{14.7^{+0.3}_{-0.4}}$ Pa s, approximately an order of magnitude higher than the canonical values used in the published literature. This result suggests that future studies should consider the effect of potentially higher interior viscosities, which will act to inhibit both convective vigor and tidal heat generation.

The largest modeled uncertainties stem from the rheology of the icy shell. These uncertainties in rheology would be mitigated by additional studies into the likely range of ice grain sizes within the convecting interior of Europa, including the effects of non-ice constituents. Further improvements could come from laboratory studies of ice rheology that improve

estimates of parameters on which viscosity depends exponentially, including the activation energies associated with various deformation mechanisms. Additionally, characterizing the thermal properties of salt–ice mixtures may improve estimates of icy shell thickness.

The largest unmodeled uncertainties are associated with the current thermal state of Europa’s icy shell. If the icy shell is in a state of rapid thickening or thinning, heat will be exchanged and stored with the changing phase of water. Because this process is unbounded, it is not considered in this study.

Future measurements of Europa that result in improbably thin or thick inferences of icy shell or layer thicknesses may be used to infer tidal dissipation within the silicate interior or that the icy shell thickness is in a state of flux. An anomalous icy shell thicknesses can be inferred as anomalous heat flux, providing key constraints on the global energy budget and non-ice material incorporation within the icy shell, and such would inform studies of geodynamic activity and potential habitability. Key contextual data is required to disentangle the potential contributions from a dynamic icy shell with those of an anomalously dissipative solid interior.

The model estimates of icy thickness and thermomechanical properties produced by this study may guide future spaceborne, landed, and subsurface exploration of Europa by defining the potential range of configurations and icy shell environments a spacecraft might encounter. This modeling approach could be extended to survey the potential structure and dynamic state of other icy ocean worlds with global subsurface oceans, including Ganymede, Callisto, Enceladus, Titan, and potentially Triton. Model outputs can additionally aid in planning for future observations of Europa’s icy shell, providing the full gamut of potential conductive, convective, porous, and total icy shell thicknesses with associated uncertainties. Future spacecraft observations by NASA’s planned Europa Clipper flagship mission and the New Frontiers–class Juno mission, as well as by ESA’s L-class JUICE mission, will prove crucial to refining these estimates.

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References

- Anderson, J. D., Schubert, G., Jacobson, R. A., et al. 1998, *Sci*, **281**, 2019
- Ashkenazy, Y. 2019, *Heliyon*, **5**, e01908
- Barnes, P. R., Mulvaney, R., Robinson, K., & Wolff, E. W. 2002, *AnGla*, **35**, 559
- Barr, A. C., & McKinnon, W. B. 2007, *JGRE*, **112**, E02012
- Barr, A. C., & Showman, A. P. 2009, in Europa, ed. R. T. Pappalardo et al. (Tucson, AZ: Univ. Arizona Press), 405
- Beuthe, M. 2016, *Icar*, **280**, 278
- Bierhaus, E. B., Zahnle, K., Chapman, C. R., et al. 2009, in Europa, ed. R. T. Pappalardo et al. (Tucson, AZ: Univ. Arizona Press), 161
- Billings, S. E., & Kattenhorn, S. A. 2005, *Icar*, **177**, 397
- Blankenship, D. D., Young, D. A., Moore, W. B., & Moore, J. C. 2009, in Europa, ed. R. T. Pappalardo (Tucson, AZ: Univ. of Ariz. Press), 631
- Bolton, S., Lunine, J., Stevenson, D., et al. 2017, *SSRv*, **213**, 5
- Buffo, J. J., Schmidt, B. E., Huber, C., & Walker, C. C. 2020, *JGRE*, **125**, e06394
- Běhouňková, M., Tobie, G., Choblet, G., et al. 2020, *GeoRL*, **48**, e90077
- Carr, M. H., Belton, M. J. S., Chapman, C. R., et al. 1998, *Natur*, **391**, 363
- Casajus, L. G., Zannoni, M., Modenini, D., et al. 2020, *Icar*, **358**, 114187
- Cox, G. F., & Weeks, W. F. 1975, Brine drainage and initial salt entrapment in sodium chloride ice (Hanover, NH: Cold Regions Research and Engineering Laboratory), <https://hdl.handle.net/11681/5820>
- De La Chapelle, S., Castelnau, O., Lipenkov, V., & Duval, P. 1998, *JGR*, **103**, 5091
- Di Paolo, F., Lauro, S. E., Castelletti, D., et al. 2016, *IJSTA*, **10**, 118
- Doggett, T., Greeley, R., Figueredo, P., & Tanaka, K. 2009, in Europa, ed. R. T. Pappalardo et al. (Tucson, AZ: Univ. Arizona Press), 137
- Fletcher, N. H. 2009, *The Chemical Physics of Ice* (Cambridge: Cambridge Univ. Press)
- Frost, H. J., & Ashby, M. F. 1982, *Deformation Mechanism Maps: The Plasticity and Creep of Metals and Ceramics* (Oxford: Pergamon)
- Gammon, P. H., Kieffe, H., Clouter, M. J., & Denner, W. W. 1983, *JGla*, **29**, 433
- Goldsby, D. L., & Kohlstedt, D. L. 2001, *JGR*, **106**, 11017
- Grasset, O., Dougherty, M., Coustenis, A., et al. 2013, *P&SS*, **78**, 1
- Green, A., Montesi, L., & Cooper, C. 2021, *JGRE*, **126**, e06677
- Greenberg, R., Hoppa, G. V., Tufts, B. R., et al. 1999, *Icar*, **141**, 263
- Hammond, N. P. 2020, *JGRE*, **125**, e06170
- Hand, K. P., Carlson, R. W., & Chyba, C. F. 2007, *AsBio*, **7**, 1006
- Hand, K. P., & Chyba, C. F. 2007, *Icar*, **189**, 424
- Hand, K. P., Murray, A., Garvin, J., et al. 2017, Europa Lander Study 2016 Report JPL D-97667, NASA JPL, <https://europa.nasa.gov/resources/58/europa-lander-study-2016-report/>
- Hauke, J., & Kossowski, T. 2011, *Quaestiones Geographicae*, **30**, 87
- Hay, H. C. F. C., & Matsuyama, I. 2019, *Icar*, **319**, 68
- Hendrix, A. R., Hurford, T. A., Barge, L. M., et al. 2019, *AsBio*, **19**, 1
- Hondoh, T. 2019, *Crystals*, **9**, 386
- Howell, S. M. 2021, Europa Monte Carlo Software, Zenodo, doi:10.5281/zenodo.4673164
- Howell, S. M., & Pappalardo, R. T. 2018, *GeoRL*, **45**, 4701
- Howell, S. M., & Pappalardo, R. T. 2019, *Icar*, **322**, 69
- Howell, S. M., & Pappalardo, R. T. 2020, *NatCo*, **11**, 1311
- Howell, S. M., Stone, W. C., Craft, K., et al. 2020, arXiv:2006.15803
- Hussmann, H., Choblet, G., Lainey, V., et al. 2010, *SSRv*, **153**, 317
- Hussmann, H., Shoji, D., Steinbrügge, G., Stark, A., & Sohl, F. 2016, *CeMDA*, **126**, 131
- Johnson, B. C., Sheppard, R. Y., Pascuzzo, A. C., Fisher, E. A., & Wiggins, S. E. 2017, *JGRE*, **122**, 2765
- JPL 2021, JPL HORIZONS Ephemerides Database (Pasadena, CA: NASA JPL, Caltech), <https://ssd.jpl.nasa.gov/horizons.cgi>
- Kalousová, K., Schroeder, D. M., & Soderlund, K. M. 2017, *JGRE*, **122**, 524
- Kalousová, K., & Sotin, C. 2020, *GeoRL*, **47**, e87481
- Kattenhorn, S. A., & Prockter, L. M. 2014, *NatGe*, **7**, 762
- Kumar, N., Hirschey, J., LaClair, T. J., Gluesenkamp, K. R., & Graham, S. 2019, *Journal of Energy Storage*, **24**, 100794
- Kuskov, O. L., & Kronrod, V. A. 2005, *Icar*, **177**, 550
- Lainey, V., Arlot, J.-E., Karatekin, O., & Van Hoolst, T. 2009, *Natur*, **459**, 957
- Leonard, E. J., Yin, A., & Pappalardo, R. T. 2020, *Icar*, **343**, 113709
- McCord, T. B., Hansen, G. B., Matson, D. L., et al. 1999, *JGRE*, **104**, 11827
- McKinnon, W. B. 1999, *GeoRL*, **26**, 951
- Mitri, G., & Showman, A. P. 2005, *Icar*, **177**, 447
- Moore, W. B., Hussmann, H., Pappalardo, R., McKinnon, W., & Khurana, K. 2009, in Europa, ed. R. T. Pappalardo et al. (Tucson, AZ: Univ. Arizona Press), 369
- Moresi, L.-N., & Solomatov, V. 1995, *PhFl*, **7**, 2154
- Nimmo, F. 2004, *JGRE*, **109**, E01003
- Nimmo, F., & Gaidos, E. 2002, *JGRE*, **107**, 5021
- Nimmo, F., & Manga, M. 2008, in Europa, ed. R. T. Pappalardo et al. (Tucson, AZ: Univ. Arizona Press), 589
- Nimmo, F., Pappalardo, R. T., & Giese, B. 2003, *Icar*, **166**, 21
- Nimmo, F., & Schenk, P. 2006, *JSG*, **28**, 2194
- Nimmo, F., Thomas, P., Pappalardo, R., & Moore, W. 2007, *Icar*, **191**, 183
- O'Brien, D. P., Geissler, P., & Greenberg, R. 2002, *Icar*, **156**, 152
- Ojakangas, G. W., & Stevenson, D. J. 1989, *Icar*, **81**, 220
- Panning, M. P., Stähler, S. C., Huang, H.-H., et al. 2018, *JGRE*, **123**, 163
- Pappalardo, R. T., Head, J. W., Greeley, R., et al. 1998, *Natur*, **391**, 365
- Press, W. H., & Teukolsky, S. A. 1990, *ComPh*, **4**, 669
- Prieto-Ballesteros, O., & Kargel, J. S. 2005, *Icar*, **173**, 212
- Quick, L. C., & Marsh, B. D. 2015, *Icar*, **253**, 16
- Ramseier, R. O. 1967, Self-diffusion in ice monocrystals CRREL, Res. Rep. 232
- Segatz, M., Spohn, T., Ross, M., & Schubert, G. 1988, *Icar*, **75**, 187
- Showman, L., & Han, A. P. 2004, *JGRE*, **109**, E01010
- Sinha, N. K. 1989, *Cold Regions Science and Technology*, **17**, 127
- Sotin, C., Tobie, G., Wahr, J., & McKinnon, W. 2009, Europa (Tucson, AZ: Univ. Arizona Press), 85
- Spencer, J. R., Tamppari, L. K., Martin, T. Z., & Travis, L. D. 1999, *Sci*, **284**, 1514
- Spohn, T., & Schubert, G. 2003, *Icar*, **161**, 456
- Stähler, S. C., Panning, M. P., Vance, S. D., et al. 2018, *JGRE*, **123**, 206
- Tobie, G., Choblet, G., & Sotin, C. 2003, *JGRE*, **108**, 5124
- Tobie, G., Mocquet, A., & Sotin, C. 2005, *Icar*, **177**, 534
- Turcotte, D., & Schubert, G. 2014, *Geodynamics* (Cambridge: Cambridge Univ. Press), 411
- Urquhart, A., & Bauer, S. 2015, *International Journal of Rock Mechanics and Mining Sciences*, **78**, 350
- Vilella, K., Choblet, G., Tsao, W.-E., & Deschamps, F. 2020, *JGRE*, **125**, e06248
- Zahnle, K., Schenk, P., Levison, H., & Dones, L. 2003, *Icar*, **163**, 263
- Zolotov, M. Y., & Shock, E. L. 2001, *JGR*, **106**, 32815